

S. M. MEHRAB HOSSAIN JAYEED

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SUMMARY

Final-year Computer Science undergraduate with a strong academic record and practical experience in machine learning, LLM integration, and full-stack development. Passionate about frontier technologies including predictive AI systems, large language models, and integrative AI pipelines. Experienced in building real-world applications that combine intelligent automation, structured data workflows, and modern web and mobile technologies. Recognised for engineering excellence through competitive project showcasing and ML competition performance.

TECHNICAL SKILLS

- **Machine Learning & AI:** Supervised learning, LSTM time-series forecasting, predictive analytics, NLP fundamentals, model evaluation, RandomizedSearchCV, scikit-learn
- **LLMs & AI Integration:** Groq API, Gemini API, prompt engineering, RAG pipelines, LLM-assisted workflows, Anthropic MCP
- **Data Processing:** Feature engineering, IQR outlier removal, imputation, normalization, JSON/CSV pipelines
- **Languages:** Python, JavaScript, TypeScript, Dart, C, C++
- **Web & Mobile:** Next.js, React, Node.js, Express.js, Flutter, Tailwind CSS, Vite
- **Databases & Backend:** Supabase (PostgreSQL), MongoDB, MySQL, Firebase
- **Tools:** Git, GitHub, Chrome Extensions (Manifest V3), Postman, VS Code, Vercel

PROJECTS

Focusnyx — Student Life OS & Cognitive Shield

Next.js, Express.js, Supabase, Groq/Gemini API

[GitHub](#) | [Live Demo](#)

- Built a full-stack student productivity ecosystem combining a Next.js PWA, a Chrome extension (Manifest V3), and a Windows native companion app for distraction-free study sessions.
- Implemented a two-way real-time focus-lock sync across all three clients, browser-level URL blocking, and low-level Win32 keyboard/window hooks for desktop lockdown.
- Integrated an AI Behavioral Coach and bilingual (English/Bangla) chatbot using Groq/Gemini, an AI quiz generator from notes, and CGPA/SGPA momentum tracking with a Supabase/PostgreSQL backend.

UniShareSync — University Collaboration & Campus Management App

Flutter, Supabase, FCM, Groq API

[GitHub](#) | [Live Demo](#)

- Built a cross-platform Flutter app unifying academic resources, project collaboration, campus logistics, and peer commerce into a single role-based (Student/Faculty/Admin) ecosystem.
- Integrated a Groq-powered AI Campus Assistant with RAG-based, resource-grounded answering, plus a real-time collaborative Kanban board and whiteboard via Supabase Realtime.
- Implemented live GPS bus tracking with OpenStreetMap, a trust-scored CampusShare marketplace, and FCM push notifications across the platform.

- CS50x final project: a campus-only marketplace for students to find trusted peers for quick tasks, with ID verification and real-time chat via Socket.IO.
- Built with Flask, SQLite, and Jinja; implemented a ghosting detection system to maintain platform accountability.

COMPETITIONS & ACHIEVEMENTS

- **Datathon — ML Contest 2026** *Machine Learning Community, SMUCT*
Competed in the ML Contest by the CSE & CSIT Department at Shanto-Mariam University. Built an end-to-end pipeline on a loan approval dataset: imputation, IQR outlier removal, feature encoding, normalization, and 20+ models with RandomizedSearchCV tuning.
- **2nd Place — Software Project Showcase 2026** *Software Community, SMUCT*
Secured 2nd place at the inter-department Software Project Showcase 2026 organised by the Software Community.
- **Champion — Project Showcasing 2025** *Robotics & IoT Community, SMUCT*
Team X recognised as champion, demonstrating engineering and problem-solving skills in a competitive academic setting.

CERTIFICATIONS

- **CS50's Introduction to Artificial Intelligence with Python** — HarvardX, edX Verified Certificate
Covered search, optimization, machine learning, neural networks, and NLP.
- **CS50x: Introduction to Computer Science** — HarvardX, edX Verified Certificate
Built GigCampus as the final project using Flask, SQLite, and Socket.IO.
- **Anthropic Model Context Protocol (MCP) Introduction** — Anthropic
Studied how frontier AI systems expose tool-use capabilities to external agents and integrative pipelines.
- **AWS Cloud Technical Essentials** — Grameenphone Academy

EDUCATION

Bachelor of Science in Computer Science & Engineering <i>Shanto-Mariam University of Creative Technology</i> Current Semester: 9th CGPA: 3.93 / 4.00 (up to 8th semester)	Expected Graduation: 2026 <i>Dhaka, Bangladesh</i>
Higher Secondary Certificate (HSC) <i>Dhaka City College</i> Science Group GPA: 5.00 / 5.00	2021 <i>Dhaka, Bangladesh</i>
Secondary School Certificate (SSC) <i>Armanitola Govt. High School</i> Science Group GPA: 5.00 / 5.00	2019 <i>Dhaka, Bangladesh</i>

LANGUAGES

- Bangla: Native | English: Upper-Intermediate (B2)